

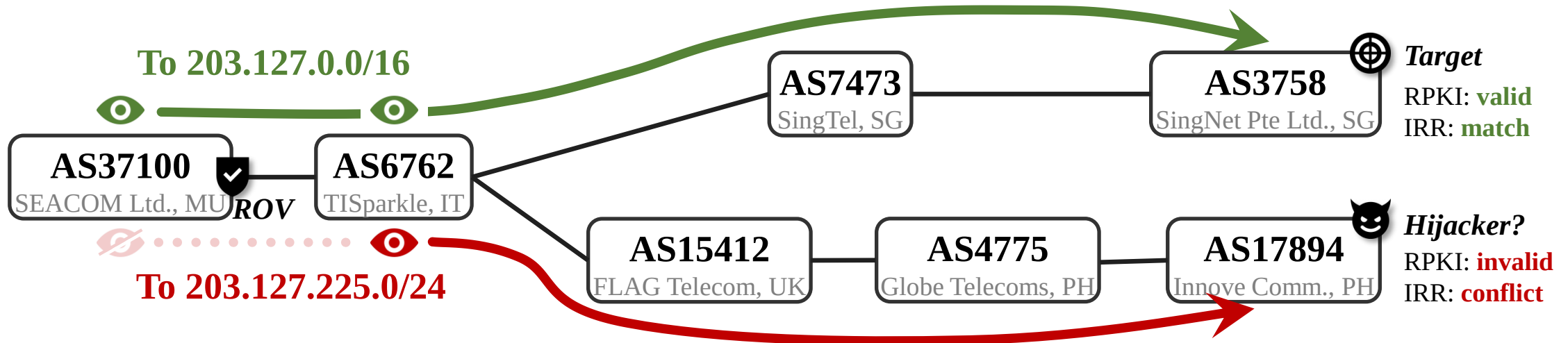
Understanding the Stealthy BGP Hijacking Risks in the ROV Era

Yihao Chen, Qi Li, Ke Xu, Zhuotao Liu, Jianping Wu
Tsinghua University

Content

1. BGP Hijacking in the ROV Era: A Real-World Case
2. Stealthy BGP Hijacking
3. Empirical Analysis & Insights
4. Simulation-Based Risk Evaluation
5. Discussion & Future Work

BGP Hijacking in the ROV Era: A Real-World Case



* First observed: 2023-10-01 Recently observed: 2025-02-05

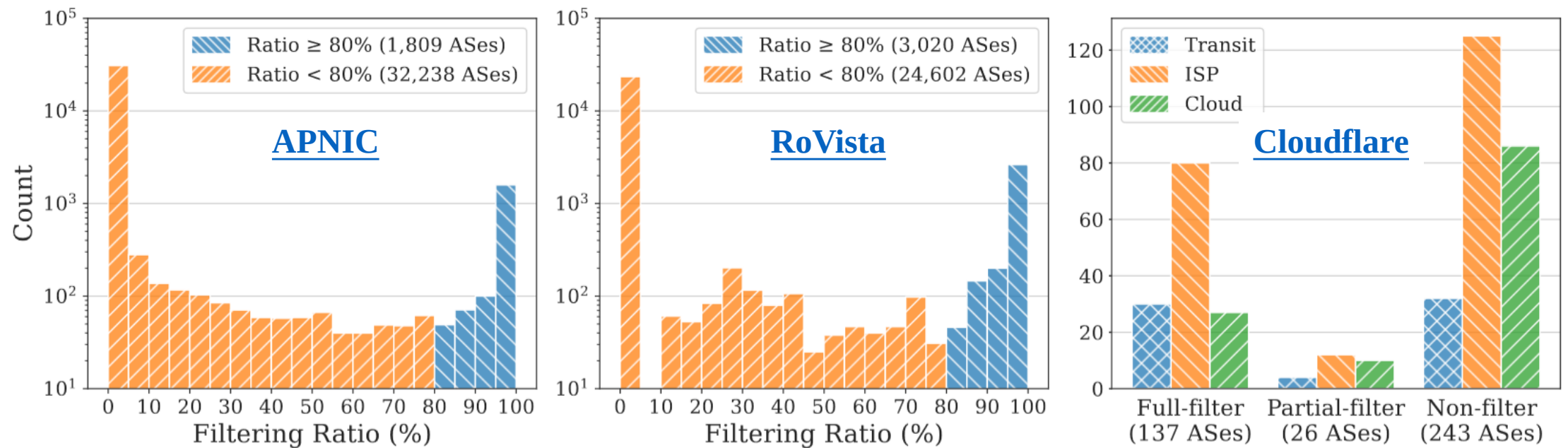
* While in the form of BGP hijacking, this case is very likely to be caused by misconfiguration; we awaiting response from relevant operators.

- AS17894 mis-announces a /24 without RPKI or IRR authorization, while AS3758 is the legitimate origin
- ROV-enabled AS37100 discards the invalid route, and thus **has the legitimate /16 route only**
 - By seeing routing tables, AS37100 and its customers are **not aware of the potential hijacking attempt**, and tend to believe their traffic be forwarded correctly.
- However, **legacy AS6762 accepts the illegitimate route**, actually forwarding /24 traffic to AS17894

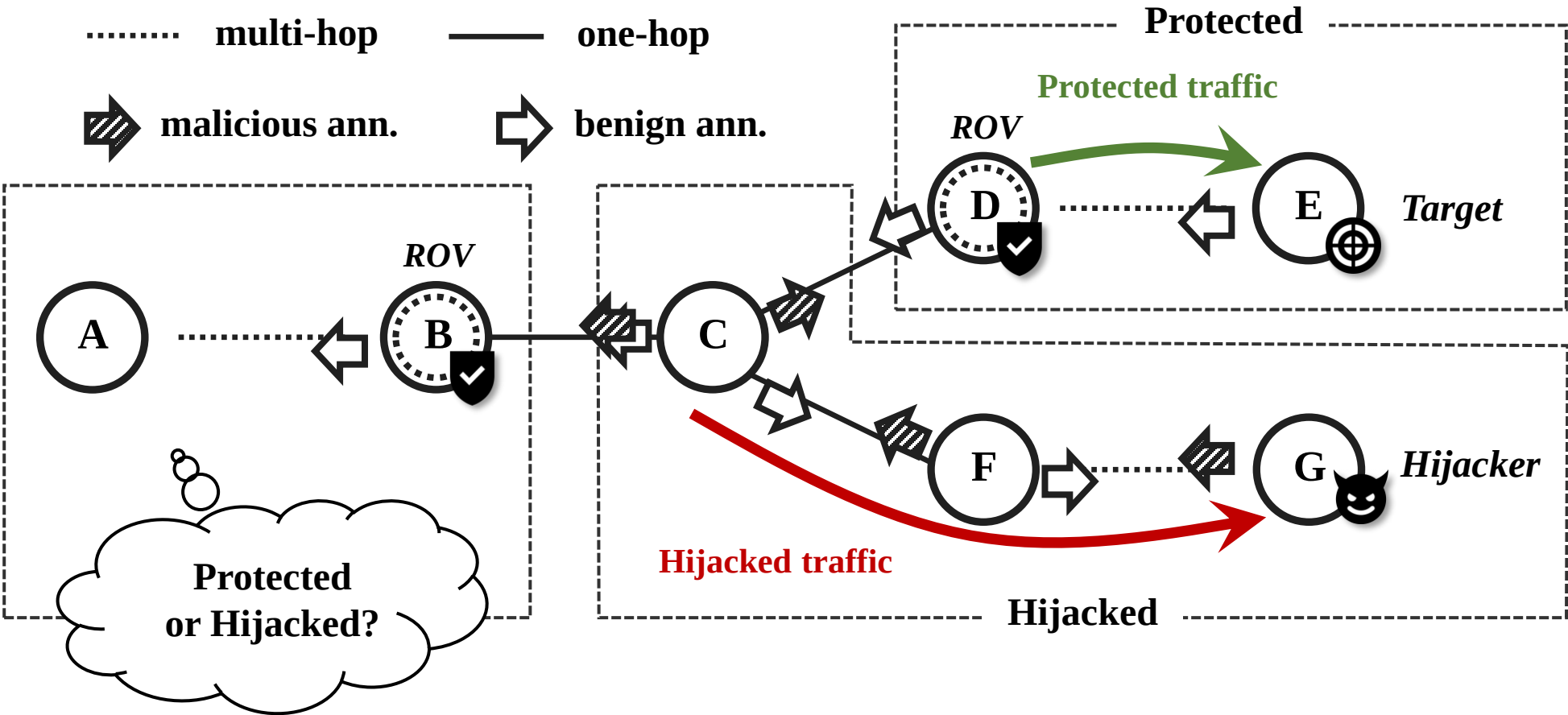
BGP hijacking still succeeds, and becomes less spotted (stealthier), due to the incomplete ROV deployment in the current Internet?

RPKI/ROV in Today's Internet

- ROAs have covered **53.9% IPv4 prefixes** and **59.2% IPv6 prefixes***
- Yet ROV deployment is **presumably limited**

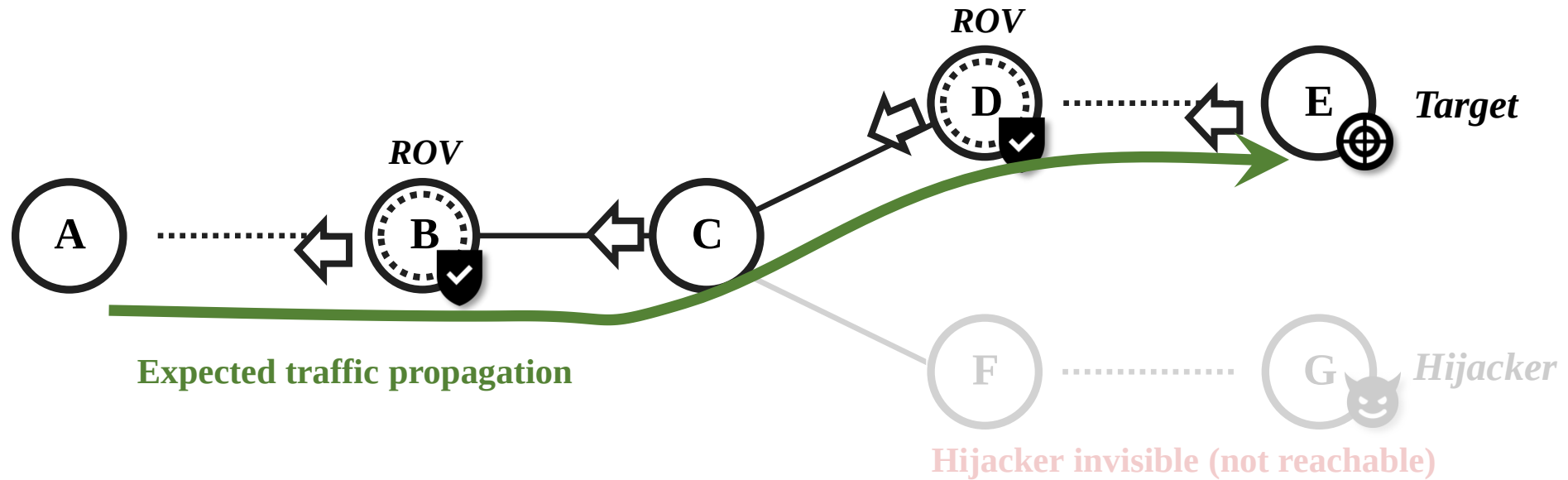


BGP Hijacking Under Partial ROV deployment



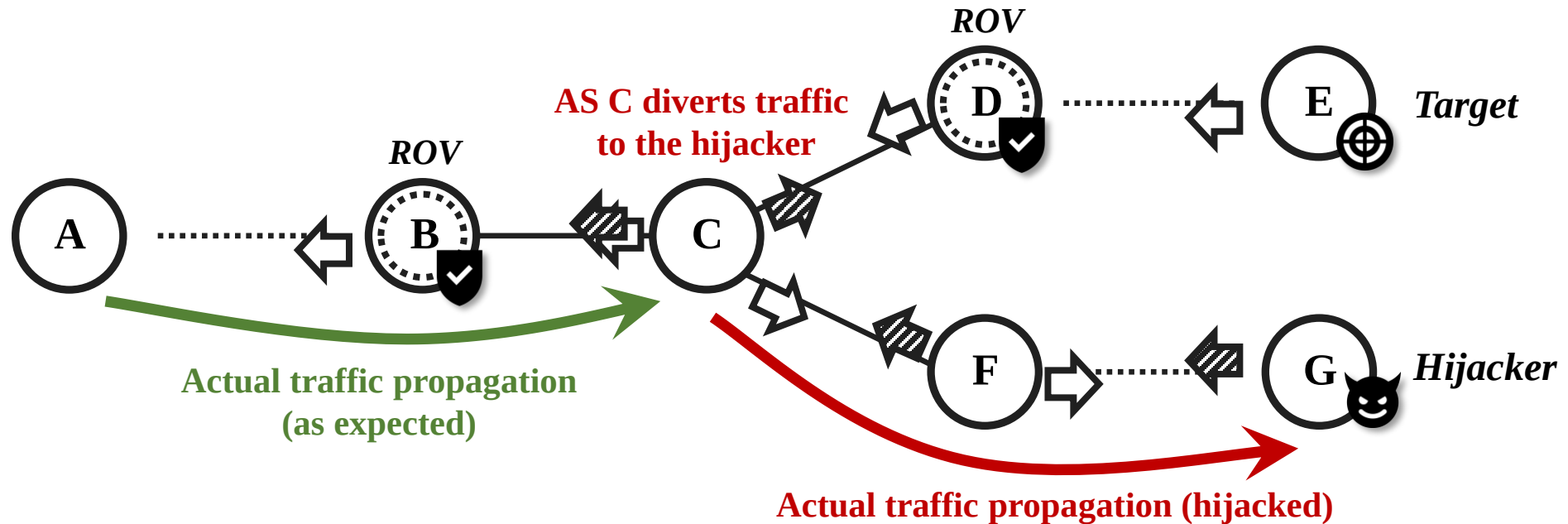
* This is the general topology structure that illustrates how BGP hijacking could succeed despite partial ROV deployment. It does not reflect the whole Inter-AS topology, but only the part that makes the stealthy hijacking scenario possible.

AS A's Perspective



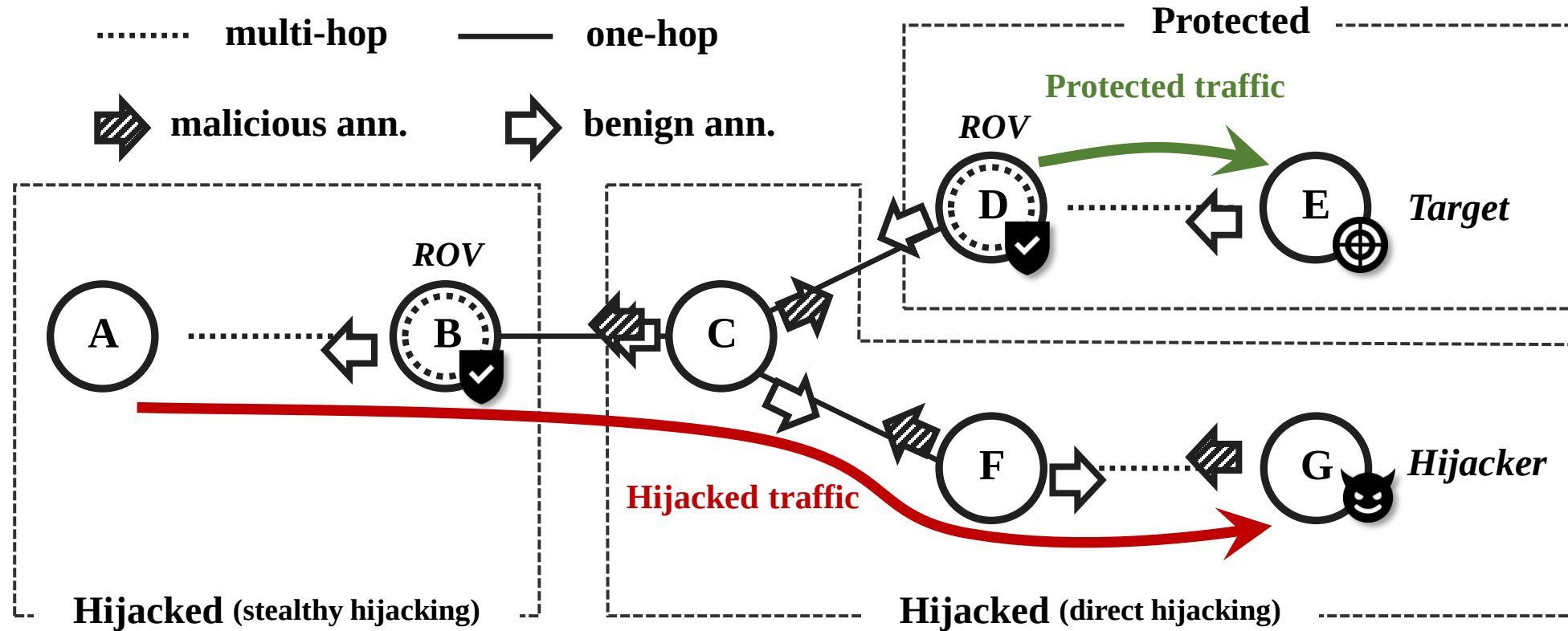
- The hijacker is invisible to AS A on the control plane, due to the ROV filtering of AS B
- AS A would expect its traffic safely forwarded along the correct path (green)

Global Perspective



- AS C, however, is not protected by ROV, and diverts traffic to the hijacker, as long as it accepts the prefix (either the target prefix or a subprefix of it) announced by the hijacker.
- AS A falls victim to hijacking unknowingly.

Stealthy Hijacking: An Unexpected Consequence



BGP hijacking that is invisible from the victim on the control plane

Empirical Analysis of Stealthy Hijacking in the Wild

Data Sources

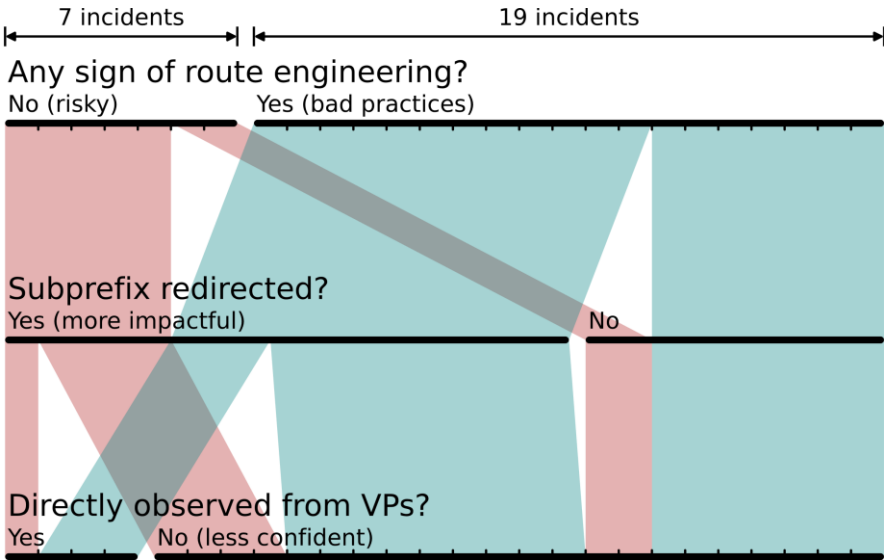
- RIB data from RouteViews collectors (route-views2, amsix, wide) on Oct 1, 2023 (56M routes, 86 vantage points).
- Validation: RIPE NCC RPKI, RADB IRR, WHOIS from 5 RIRs, CAIDA AS Organization dataset..

Results

- **26 incidents:** 19 linked to poor route engineering (e.g., outdated RPKI/IRR), 7 potential hijacks.
- **Affected:** 1,205 routes, 40 prefixes, 27 origins.

Type	#Prefixes	#Origins	#Routes	#VPs
Risky incidents	10	7	120	45
Bad practices	30	20	1,085	57
Total	40	27	1,205	57

#0 Most wild stealthy hijacking is caused by bad route engineering practice.



Empirical Analysis: Public Service

Our website to publish stealthy hijacking incidents in the wild:

<https://yhchen.cn/stealthy-bgp-hijacking/>

- Set up search options and date range
- Search your interested ASN/prefix/organization
- Review the results and feedback
- Beta feature: GPT analysis with routing knowledge augmentation

Incident #892

Category: Potential Stealthy Hijacking

Time: 2025-02-05 12:00

Affected prefixes: 203.127.0.0/16, 203.127.225.0/24

Expected origins: AS3758 (SingNet Pte Ltd, SG)

Unexpected origins: AS17894 (Innovate Communications, PH)

GPT Analysis (beta)

This incident suggests potential stealthy BGP hijacking due to inconsistent routing for 203.127.225.0/24, a sub-prefix of 203.127.0.0/16. Legitimate origin AS3758 (SingNet, SG) is authorized by RPKI ROAs (AS3758 and AS19905) to announce up to /24 sub-prefixes under 203.127.0.0/16. However, AS17894 (Innovate Communications, PH) is mis-announcing 203.127.225.0/24 without RPKI authorization. ROV-enabled ASes (e.g., AS37100 with 100% ROV filtering) discard the invalid route, but legacy AS6762 (SEABONE-NET, IT, 0% ROV) accepts it. Traffic from VP 267613 and AS6762 itself (tagged "Direct VP View") is diverted at AS6762 via path [15412 4775 17894], bypassing the legitimate path [6461 7473 3758]. AS37100, protected by ROV, observes only the legitimate route, creating a stealthy diversion: traffic from ROV-enabled ASes reaches AS3758, while traffic traversing AS6762 is redirected to AS17894.

The hijacking leverages partial ROV deployment. AS6762, a transit AS without ROV, propagates the invalid route. The /24 sub-prefix triggers longest-prefix matching, overriding the legitimate /16 route at AS6762. Critically, AS3758 (legitimate) does not appear in the unexpected route, and AS17894 (PH) and AS3758 (SG) are in different countries (tagged "Different Countries"), with no known business relationship per CAIDA data.

Stealthy BGP Hijacking Incidents

Download About Contact

Your ASN/Prefix/Organization

Advanced options

showing all 907 results.

Expected origins: AS9900 (Singapore Telecommunications Ltd, Magix Services, SG)

Unexpected origins: AS64013 (CONA HOSTING SDN BHD, KR)

#905

2025-02-05 12:00

Category: Potential Stealthy Hijacking

Affected prefixes: 115.117.0.0/16, 115.117.192.0/18, 115.117.224.0/24

Expected origins: AS10199 (Tata Communications Limited, IN)

Unexpected origins: AS17762 (Tata Teleservices (Maharashtra) Ltd, IN)

#904

2025-02-05 12:00

Category: Bad Operational Practice

Affected prefixes: 201.49.228.0/22

Expected origins: AS52532 (Speednet Telecomunicações Ltda ME, BR)

Unexpected origins: AS5 (WFA Group LLC, US)

#903

2025-02-05 12:00

Category: Bad Operational Practice

Affected prefixes: 199.59.94.0/24, 99.192.207.0/24

Expected origins: AS42567 (MOJOHOST B.V., NL)

Unexpected origins: AS27589 (MOJOHOST, US)

#902

2025-02-05 12:00

Category: Potential Stealthy Hijacking

Affected prefixes: 154.214.224.0/19, 154.214.232.0/22

Expected origins: AS328608 (Africa on Cloud, ZA)

Unexpected origins: AS62387 (meerfarbig GmbH & Co. KG, DE)

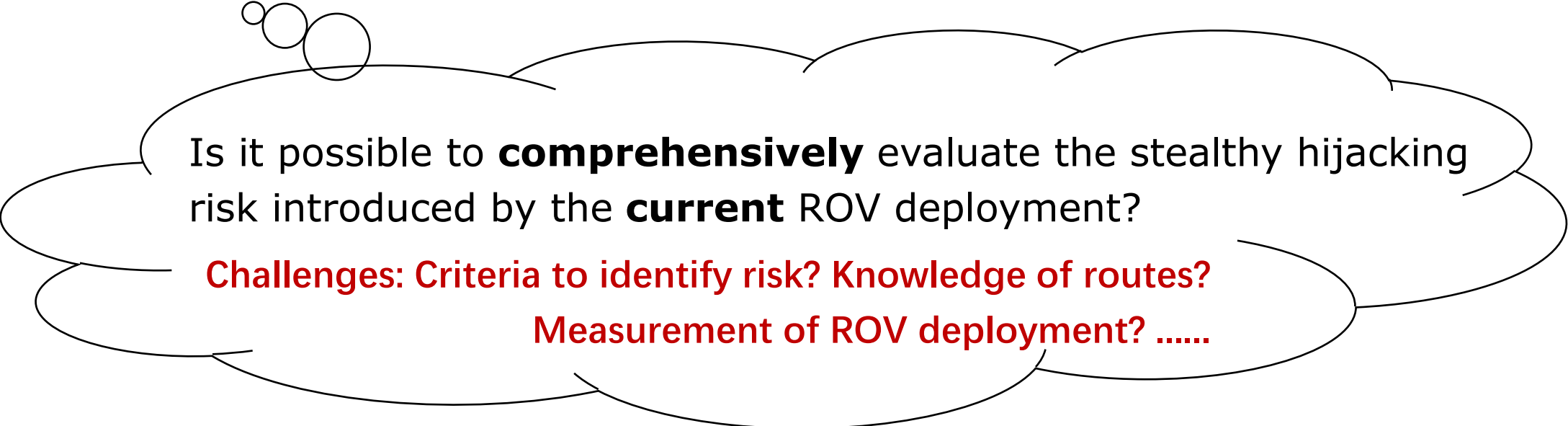
* Daily analysis results since 2025-01-01

Simulation-Based Risk Evaluation

Questions unanswered:

The prevalence, extent, distribution, influencing factors, etc. of the risk within our real-world Internet.

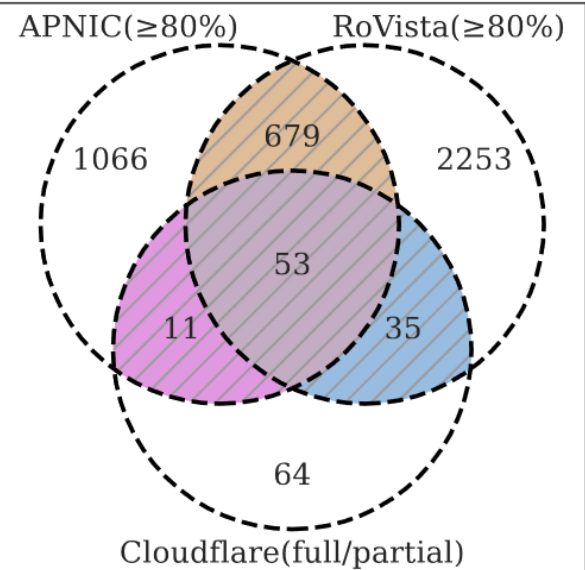
- Use **CAIDA AS relationships** to reconstruct Internet topology
- Use **APNIC, RoVista, Cloudflare** measurement to gain ROV deployment status
- Perform BGP simulation on the topology and generate **all Internet-scale routes**
- Thoroughly analyze **all possible “victim-target-hijacker” 3-tuple** to determine those at risk of stealthy BGP hijacking.



Is it possible to **comprehensively** evaluate the stealthy hijacking risk introduced by the **current** ROV deployment?

**Challenges: Criteria to identify risk? Knowledge of routes?
Measurement of ROV deployment?**

Results: Overall Risk Level



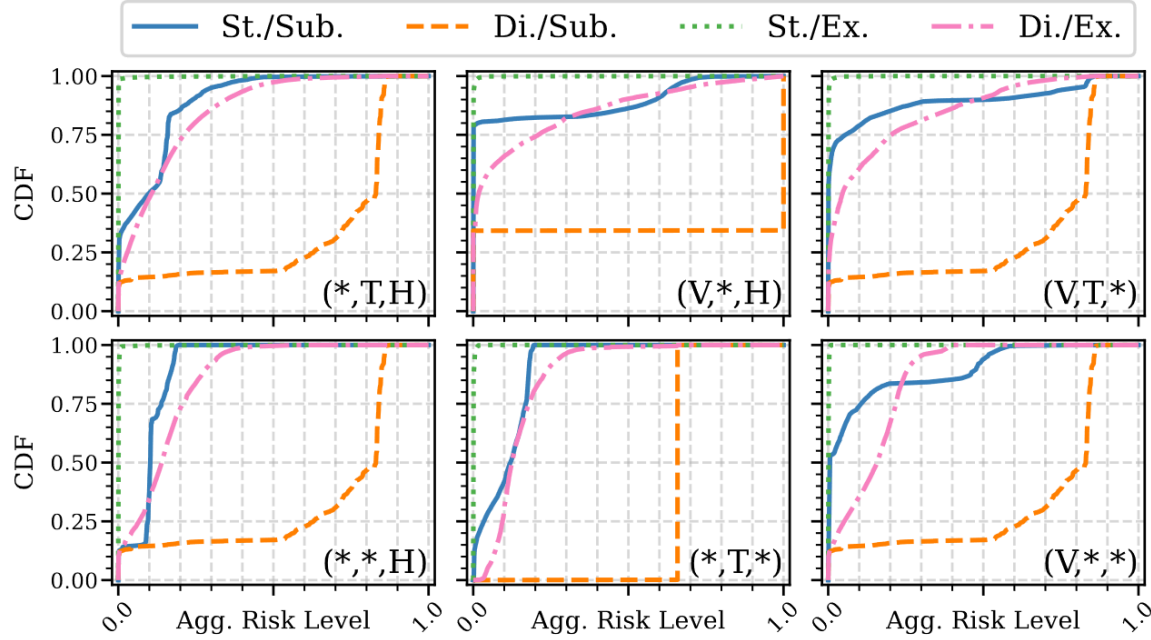
778 ASes identified ROV-enabled by at least two sources

Stealthy Hijacking	Sub-Prefix	Exact-Prefix
w/ ROV	0.104 ▲0.104	0.001 ▲0.001
w/o ROV	0.000	0.000

Direct Hijacking	Sub-Prefix	Exact-Prefix
w/ ROV	0.658 ▼0.328	0.144 ▼0.194
w/o ROV	0.986	0.338

#1 The overall success rate of stealthy hijacking increases from 0 to 10.4%, solely due to the partial ROV deployment in today’s Internet.

Results: Aggregated Risk Level



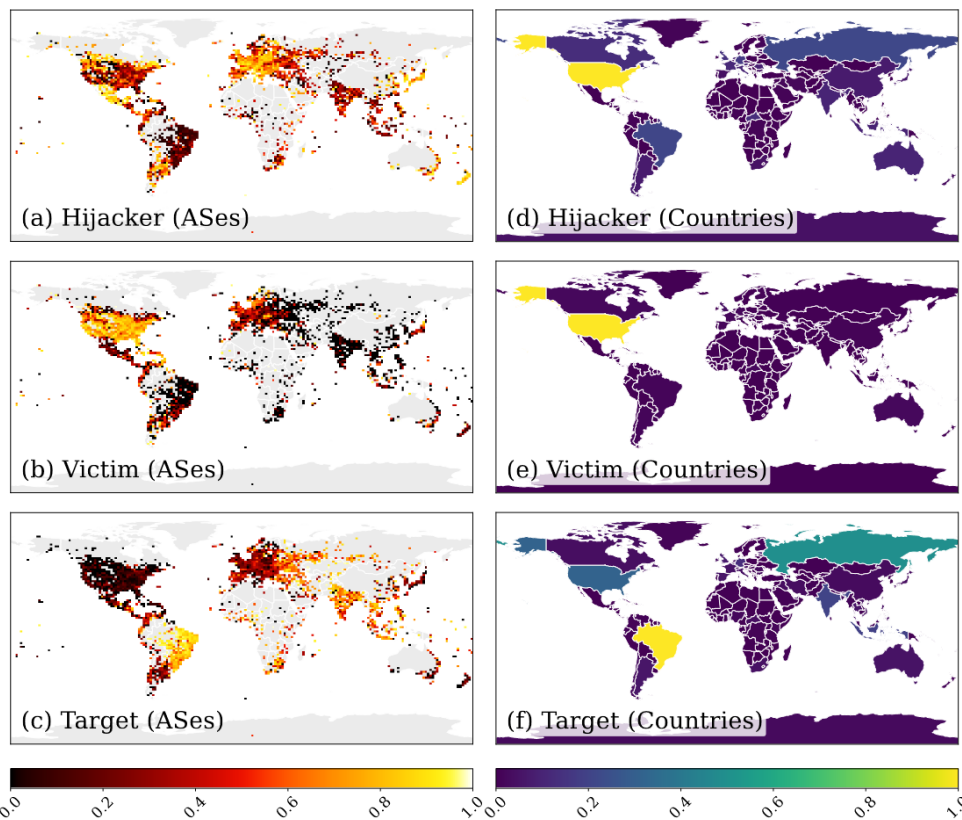
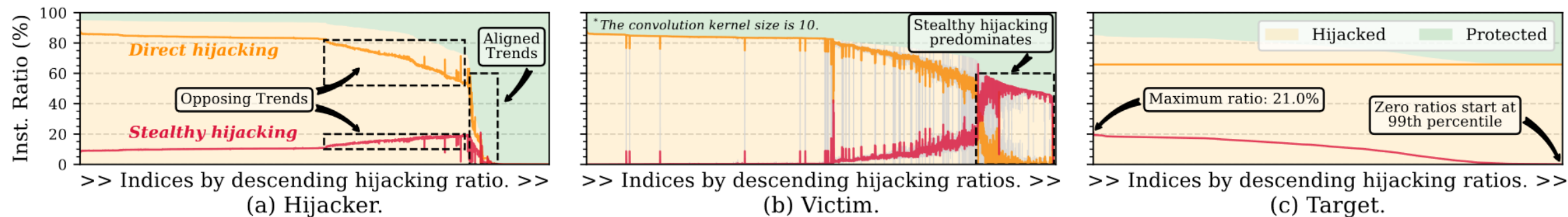
#2 Targeted stealthy hijacking has high success rates on specific AS pairs (up to 99.5%), while non-targeted one spreads risk more evenly (within 21.0%).

Statistics ¹		Hijacking Type ²			
		St./Sub.	Di./Sub.	St./Ex.	Di./Ex.
$P(*,T,H)$	min	0.000▼0.000	0.000▼0.000	0.000▼0.000	0.000▼0.000
	25th	0.003▲0.003	0.629▼0.362	0.000▼0.000	0.029▼0.132
	50th	0.096▲0.096	0.830▼0.164	0.000▼0.000	0.108▼0.176
	75th	0.159▲0.159	0.840▼0.154	0.000▼0.000	0.212▼0.280
	max	0.995▲0.995	0.880▼0.116	0.990▲0.990	0.880▼0.116
$P(V,*,H)$	min	0.000▼0.000	0.000▼0.000	0.000▼0.000	0.000▼0.000
	25th	0.000▼0.000	0.000▼1.000	0.000▼0.000	0.000▼0.116
	50th	0.000▼0.000	1.000▼0.000	0.000▼0.000	0.018▼0.252
	75th	0.000▼0.000	1.000▼0.000	0.000▼0.000	0.211▼0.307
	max	0.995▲0.995	1.000▼0.000	0.994▲0.994	1.000▼0.000
$P(V,T,*)$	min	0.000▼0.000	0.000▼0.000	0.000▼0.000	0.000▼0.000
	25th	0.000▲0.000	0.629▼0.362	0.000▼0.000	0.005▼0.061
	50th	0.000▲0.000	0.830▼0.164	0.000▲0.000	0.046▼0.208
	75th	0.055▲0.055	0.840▼0.154	0.001▲0.001	0.201▼0.382
	max	0.877▲0.877	0.880▼0.116	0.347▲0.347	0.880▼0.116
$P(*,*,H)$	min	0.000▼0.000	0.000▼0.000	0.000▼0.000	0.000▼0.000
	25th	0.096▲0.096	0.629▼0.362	0.000▲0.000	0.067▼0.172
	50th	0.103▲0.103	0.830▼0.164	0.000▲0.000	0.139▼0.184
	75th	0.139▲0.139	0.840▼0.154	0.001▲0.001	0.209▼0.220
	max	0.210▲0.210	0.880▼0.116	0.425▲0.425	0.710▼0.175
$P(*,T,*)$	min	0.000▼0.000	0.658▼0.328	0.000▼0.000	0.011▼0.011
	25th	0.034▲0.034	0.658▼0.328	0.000▲0.000	0.094▼0.122
	50th	0.120▲0.120	0.658▼0.328	0.001▲0.001	0.124▼0.183
	75th	0.170▲0.170	0.658▼0.328	0.001▲0.001	0.179▼0.248
	max	0.210▲0.210	0.658▼0.328	0.045▲0.045	0.658▼0.328
$P(V,*,*)$	min	0.000▼0.000	0.000▼0.000	0.000▼0.000	0.000▼0.000
	25th	0.002▲0.002	0.629▼0.362	0.000▲0.000	0.052▼0.247
	50th	0.005▲0.005	0.830▼0.164	0.001▲0.001	0.157▼0.174
	75th	0.106▲0.106	0.840▼0.154	0.002▲0.002	0.218▼0.164
	max	0.854▲0.854	0.880▼0.116	0.031▲0.031	0.408▼0.049
$P(*,*,*)$ —		0.104▲0.104	0.658▼0.328	0.001▲0.001	0.144▼0.194

¹ 25th, 50th, and 75th represent the respective percentiles.

² St., Di., Sub., and Ex. stand for "stealthy", "direct", "sub-prefix", and "exact-prefix". The difference (▼/▲) is based on the comparison with no AS deploying ROV.

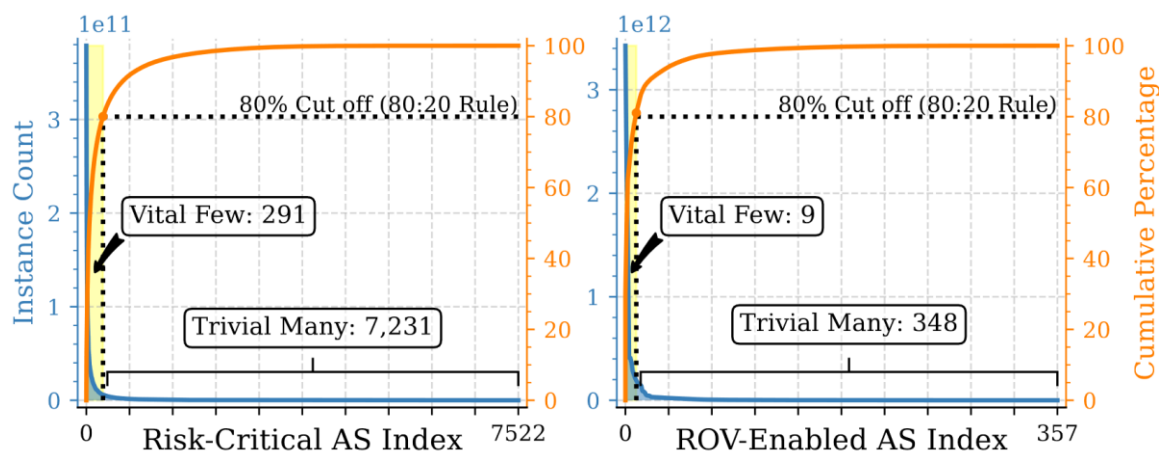
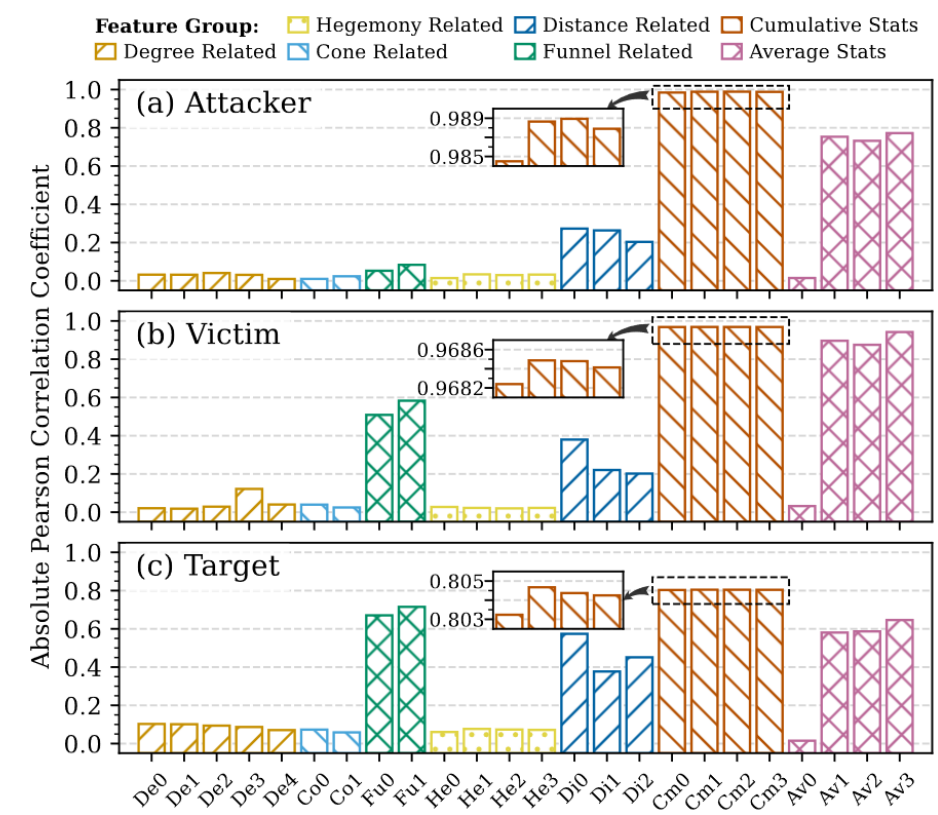
Results: Risk Distribution



#3 Stealthy hijacking risks typically follow an opposing trend to direct hijacking as ROV's impact increases.

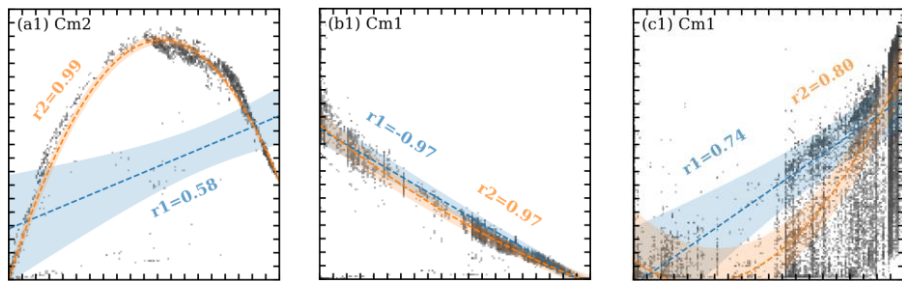
#4 Arguably, ASes most capable of hijacking are mainly in Europe and parts of North America. Victim-prone ASes are mainly in North America. Target-prone ASes are mainly in South America and South Asia.

Results: Influencing Factors



#5 Among different features, cumulative statistics of AS hegemony exhibit highest quadratic correlation with the risk.

#6 A small set of risk-critical and ROV-enabled ASes account for the majority of stealthy hijacking risk.



Discussion & Future Work

Limitations of the approach

- Prefix-specific routing simulation?
- Accuracy-coverage trade-off of ROV cross validation?
- Complex routing policy?

Potential practical application

- For ISP operation: risk assessment before peering plans
- To defend attacks: targeted probing, analysis with real-world routes,

Risk countermeasures

- Keep increasing RPKI/ROV deployment (as indicated in #3)
- Co-analyzing routing data from multiple VP to identify inconsistencies
- Collaboration among ROV-enabled ASes to share threat information
- Active blackholing on detecting stealthy hijacking (like in ROV++)

Summary

Motivation: a real-world case

BGP Hijacking in the ROV Era: A Real-World Case



- * First observed: 2023-10-01. Recently observed: 2025-02-05
- * While in the form of BGP hijacking, this case is very likely to be caused by misconfiguration; we awaiting response from relevant operators.
- AS17894 mis-announces a /24 without RPKI or IRR authorization, while AS3758 is the legitimate origin
- ROV-enabled AS37100 discards the invalid route, and thus **has the legitimate /16 route only**
 - By seeing routing tables, AS37100 and its customers are **not aware of the potential hijacking attempt**, and tend to believe their traffic be forwarded correctly.
- However, **legacy AS6762 accepts the illegitimate route**, actually forwarding /24 traffic to AS17894

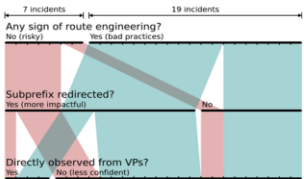
BGP hijacking still succeeds, and becomes less spotted (stealthier), due to the incomplete ROV deployment in the current Internet?

Approach: empirical + simulation-based

Empirical Analysis of Stealthy Hijacking in the Wild

- Data Sources**
- RIB data from RouteViews collectors (route-views2, amsix, wide) on Oct 1, 2023 (56M routes, 86 vantage points).
 - Validation: RIPE NCC RPKI, RADB IRR, WHOIS from 5 RIRs, CAIDA AS Organization dataset.
- Results**
- **26 incidents:** 19 linked to poor route engineering (e.g., outdated RPKI/IRR), 7 potential hijacks.
 - **Affected:** 1,205 routes, 40 prefixes, 27 origins.

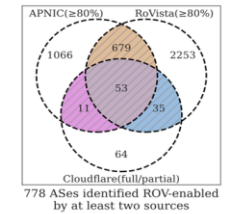
Type	#Prefixes	#Origins	#Routes	#VPs
Risky incidents	10	7	120	45
Bad practices	30	20	1,085	57
Total	40	27	1,205	57



#0 Most wild stealthy hijacking is caused by bad route engineering practice.

Evaluation: seven takeaways

Overall Risk Level



Stealthy Hijacking	Sub-Prefix	Exact-Prefix
w/ ROV	0.104 ▲0.104	0.001 ▲0.001
w/o ROV	0.000	0.000

Direct Hijacking	Sub-Prefix	Exact-Prefix
w/ ROV	0.658 ▼0.328	0.144 ▼0.194
w/o ROV	0.986	0.338

#1 The overall success rate of stealthy hijacking increases from 0 to 10.4%, solely due to the partial ROV deployment in today's Internet.

Discussion: limitations & future work

Discussion & Future Work

- Limitations of the approach**
- Prefix-specific routing simulation?
 - Accuracy-coverage trade-off of ROV cross validation?
 - Complex routing policy?
- Potential practical application**
- For ISP operation: risk assessment before peering plans
 - To defend attacks: targeted probing, analysis with real-world routes,
- Risk countermeasures**
- Keep increasing RPKI/ROV deployment (as indicated in #3)
 - Co-analyzing routing data from multiple VP to identify inconsistencies
 - Collaboration among ROV-enabled ASes to share threat information
 - Active blackholing on detecting stealthy hijacking (like in ROV++)

Thank you!

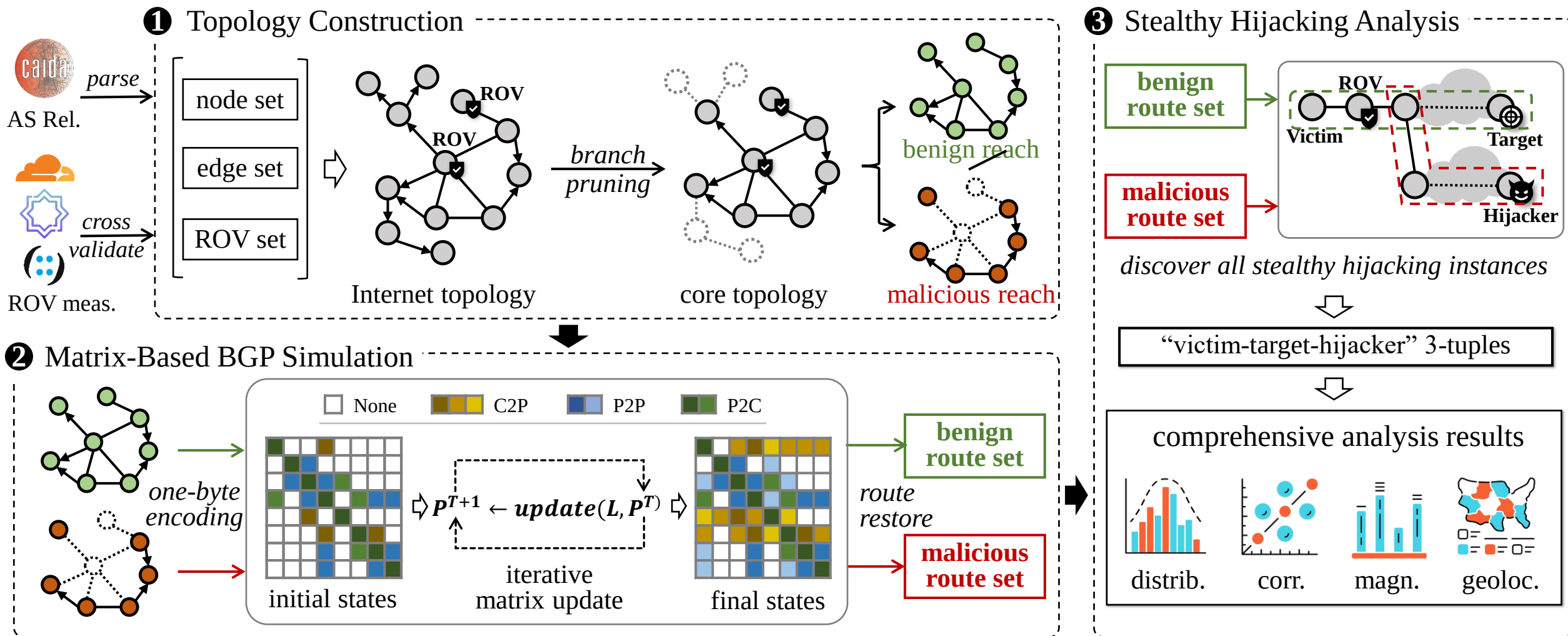
Key Takeaways

- Partial ROV deployment introduces stealthy hijacking risks to the current Internet.
- Risk evaluation can be achieved with complete knowledge of routes and ROV deployment.
- Our approach cross-validates multiple sources of ROV measurements and infers BGP routes.
- A 10.4% success rate of stealthy hijacking is introduced by the current ROV deployment.

Contact: yh-chen21@mails.tsinghua.edu.cn (Yihao Chen)

Online Service: <https://yhchen.cn/stealthy-bgp-hijacking/>

Appendix: Simulation-Based Analysis Workflow

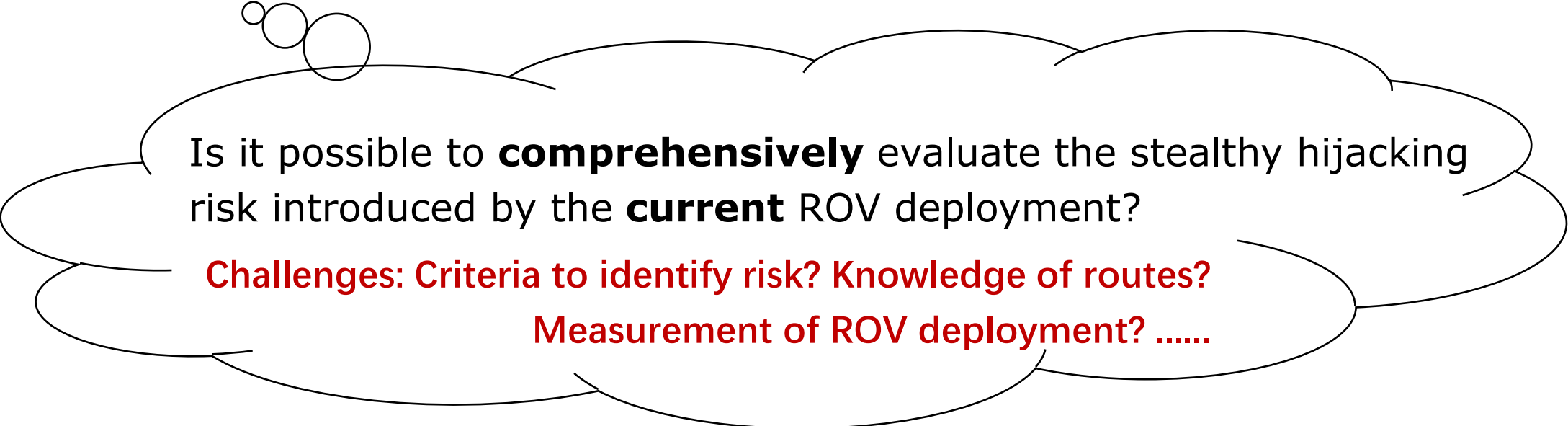


Appendix: Gap in Risk Evaluation

- ROV++ [Morillo et al, NDSS'21]
 - Assume **unrealistic ROV deployment** for experiments.
 - Evaluate with **limited stealthy hijacking samples**.
 - Present **overall hijacking rate** only.

Questions unanswered:

The prevalence, extent, distribution, influencing factors, etc. of the risk within our real-world Internet.



Is it possible to **comprehensively** evaluate the stealthy hijacking risk introduced by the **current** ROV deployment?

**Challenges: Criteria to identify risk? Knowledge of routes?
Measurement of ROV deployment?**